

Ultrafine metal nanoparticles isolated on oxide nano-islands as exceptional sintering-resistant catalysts

Received: 15 June 2024

Accepted: 9 January 2025

Published online: 10 March 2025

 Check for updates

Tao Zhou^{1,6}, Xu Li^{1,2,6}, Jiankang Zhao^{1,6}, Lei Luo¹, Yanru Wang³, Zizhen Xiao¹, Sunpei Hu¹, Ruyang Wang⁴, Zekun Zhao¹, Chengyuan Liu⁴, Wenlong Wu⁵, Hongliang Li¹, Zhirong Zhang¹, Long Zhao⁴, Han Yan¹✉ & Jie Zeng^{1,5}✉

Ultrafine nanoparticles (NPs) have attracted extensive research interest, especially in heterogeneous catalysis. However, the inherent sintering propensity of NPs has been a major obstacle to their catalytic stability. Here we report an isolation strategy to preserve highly dispersed ultrafine NPs under extremely harsh conditions. Oxide nano-islands were grafted between the catalyst support and metal NPs, serving as a general approach by following a charge attraction principle. Specifically, LaO_x nano-islands were ideally suited for stabilizing Ru NPs among the synthetic library, exhibiting strong adhesion to minimize the chemical potential and disconnect the sintering path. Thus, ultrafine Ru NPs in $\text{Ru/LaO}_x\text{-SiO}_2$ were isolated, maintaining a mean size of 1.4 nm in CO- and H_2 -rich atmosphere during efficient catalysis for methane dry reforming at 800 °C for 400 h. This isolation strategy has proved effective for many other metals on various supports, paving a practical way for the design of sintering-resistant catalysts.

Ultrafine nanoparticles (NPs), an essential constituent of nanomaterials, have exhibited intriguing properties applicable in various fields^{1–3}. Notably, benefiting from their abundant exposed surface and enhanced metal utilization, ultrafine NPs distinguish themselves in heterogeneous catalysis^{4–7}. However, the nature of high chemical potential endows ultrafine NPs with thermodynamic instability, leading to their inherent propensity for aggregation^{8,9}. During catalytic reactions, the spontaneous aggregation of NPs results in the elimination of active sites, whose kinetics depend on the competition between van der Waals attraction and electrostatic repulsion^{10–12}. The former drives the shortening of interparticle distance, while the latter creates an energy

barrier to delay aggregation before NP attachment. Once the energy barrier is surmounted, the collision of NPs sharply decreases their energy. Noteworthy, the aggregation kinetics are strongly affected by environmental parameters¹³. Under reductive atmospheres at high temperatures, the mobility of NPs is greatly intensified, while the hindrance of energy barriers on aggregation is largely weakened^{8,14}. Thus, thermally and chemically induced aggregation, commonly referred to as sintering, has posed a substantial challenge to the stability of heterogeneous catalysts^{15–18}.

Aggregation of NPs can be impeded primarily through the regulation of interparticle distance and energy barrier. To enlarge the

¹Hefei National Research Center for Physical Sciences at the Microscale, Key Laboratory of Strongly-Coupled Quantum Matter Physics of Chinese Academy of Sciences, Key Laboratory of Surface and Interface Chemistry and Energy Catalysis of Anhui Higher Education Institutes, Department of Chemical Physics, University of Science and Technology of China, Hefei, People's Republic of China. ²School of Materials and Energy, University of Electronic Science and Technology of China, Chengdu, People's Republic of China. ³Instruments Center for Physical Science, University of Science and Technology of China, Hefei, People's Republic of China. ⁴National Synchrotron Radiation Laboratory, University of Science and Technology of China, Hefei, People's Republic of China. ⁵School of Chemistry & Chemical Engineering, Anhui University of Technology, Ma'anshan, People's Republic of China. ⁶These authors contributed equally: Tao Zhou, Xu Li, Jiankang Zhao. ✉e-mail: hanyanustc@ustc.edu.cn; zengj@ustc.edu.cn

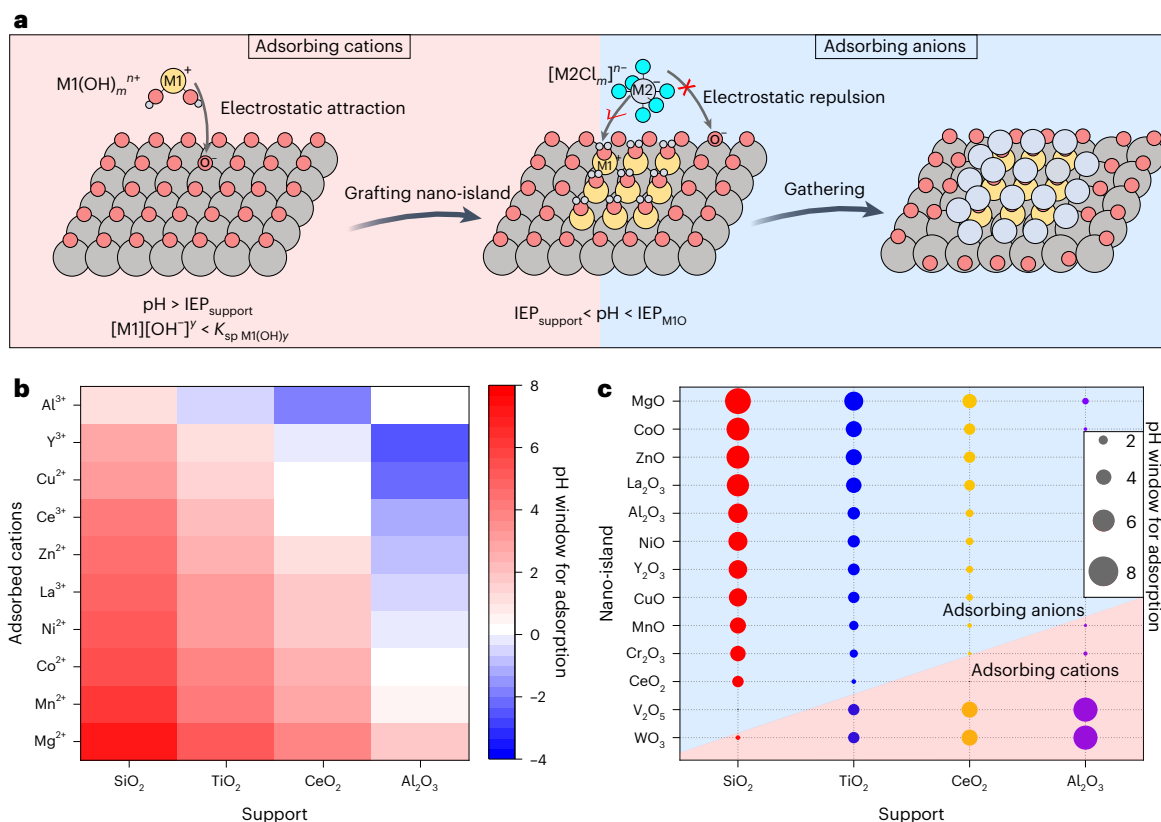


Fig. 1 | Design principle and synthetic library of nano-islands. **a**, Systematic diagrams illustrating the synthetic processes of nano-islands and gathered metals. **b**, The pH window for the adsorption of metal cations on normal supports to fabricate nano-islands. **c**, The pH window for the adsorption of ions of the second metal on each combination of nano-island and support.

interparticle distance, a solid support with a high surface area is often employed to disperse NPs, thus slowing their approach¹⁹. Furthermore, the interfacial adhesion between NPs and support lowers the chemical potential of the metal and their mobility²⁰. On the other hand, the energy barrier can be increased by coating the surface of NPs¹³. Charged or steric coverage is usually utilized to prevent the collision of NPs to a certain extent^{21–26}. However, these modified ultrafine NPs are still prone to aggregation under extremely harsh conditions, since high temperatures and reductive atmospheres strikingly accelerate their approach and collision²⁷. In comparison, an essential solution to prevent NP aggregation is the fixation of interparticle distance. Localization of metal NPs in nanoscale domains with strong adhesion can effectively reduce the chemical potential of the metal⁸. Thus, the aggregation of NPs is anticipated to be impeded or even stopped.

Herein we achieve the fabrication of exceptionally sintering-resistant metal NPs via an isolation strategy. By precisely controlling the pH during synthesis, oxide nano-islands were grafted between the catalyst support and metal NPs, serving as a universal approach for the preparation of catalysts. A synthetic library was established, enabling us to customize the ensemble structure by selecting appropriate active metals, nano-islands and catalyst support. In terms of designing sintering-resistant catalysts, LaO_x clusters were identified as the most effective nano-islands. Exceptionally sintering-resistant Ru NPs that survived in a CO - and H_2 -rich atmosphere at 800 °C were fabricated. Specifically, the strong $Ru-LaO_x$ adhesion minimized the chemical potential of Ru, resulting in the Ru NP being located preferentially on nano-islands over the SiO_2 support. Further migration or extrication of Ru NPs was inhibited by the insurmountable chemical potential gap between Ru on SiO_2 and LaO_x , referred to as a ‘potential well’ (61.6 kJ mol⁻¹). Thus, the propensity for sintering was eliminated by isolating Ru NPs on LaO_x . This outstanding sintering

resistance was utilized for catalysing dry reforming of methane (DRM, $CH_4 + CO_2 \rightarrow 2CO + 2H_2$), a promising process for syngas production but one that suffers from poor catalyst durability^{28,29}. By using the isolation strategy, ultrafine Ru NPs isolated on the SiO_2 -supported LaO_x nano-islands (Ru/LaO_x-SiO_2) still maintained a mean size of 1.4 nm even after catalysing DRM at 800 °C for 400 h. Meanwhile, the coke-free catalytic performance enabled by LaO_x -facilitated CO_2 activation guarantees high activity and selectivity, outperforming the counterparts (Ru/SiO_2 and Ru/La_2O_3). This adaptable isolation strategy has proved universal for many other metals on various supports, showing substantial perspectives in designing stabilized catalysts for numerous catalytic transformations.

Design principle of nano-island structure

Creating anisotropy on the catalyst support is crucial for blocking the sintering path of NPs. Here, we propose an approach of fabricating heterogeneous oxide nano-islands, where the synthetic craft lies in manipulating charge attraction. As shown in Fig. 1a, the surface of the oxide support carries a negative charge when the pH exceeds its isoelectric point (IEP), whereas excessive pH value causes the precipitation of metal hydroxides. Thus, metal cations of nano-islands should be adsorbed within the pH window, with one end reaching the IEP of the support and the other at the pH corresponding to the solubility product constant (K_{sp}) of metal hydroxide. Wider pH windows benefit the higher adsorption efficiency of metal cations. This pH-controlled strong electrostatic adsorption method ensures the uniform dispersion of metal species on the support, thus forming oxide nano-islands^{30,31}. Subsequently, a negatively charged support and positively charged nano-islands are obtained by adjusting the solution pH between their IEPs. The matched electrostatic attraction drives anions of the second metal to gather around the nano-islands

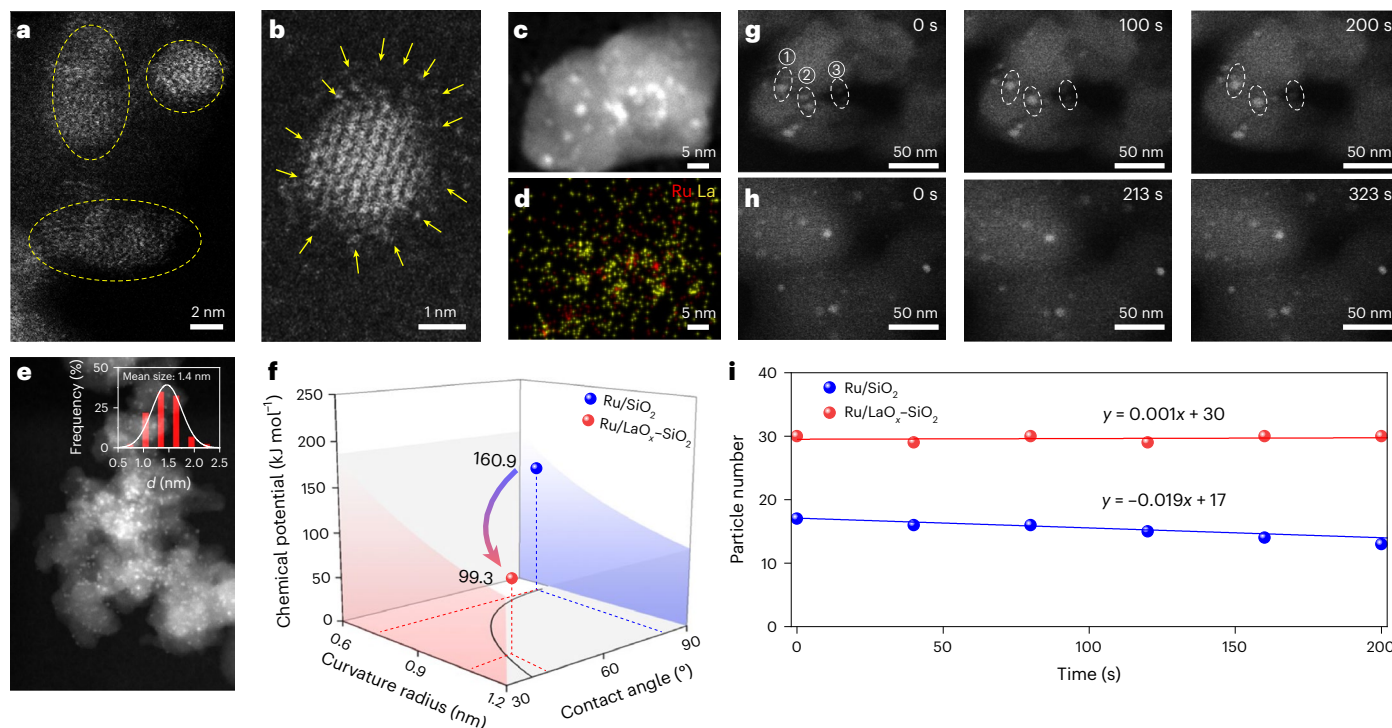


Fig. 2 | Electron microscopy characterizations. **a, b**, Atomic-resolution HAADF-STEM images of $\text{LaO}_x\text{-SiO}_2$ (**a**) and $\text{Ru/LaO}_x\text{-SiO}_2$ (**b**). **c, d**, HAADF-STEM image (**c**) and corresponding energy dispersive X-ray elemental mapping (**d**) of $\text{Ru/LaO}_x\text{-SiO}_2$. **e**, HAADF-STEM image and corresponding size distribution of

$\text{Ru/LaO}_x\text{-SiO}_2$. **f**, Chemical potential as a function of curvature radius. **g, h**, Sequential images of Ru/SiO_2 (**g**) and $\text{Ru/LaO}_x\text{-SiO}_2$ (**h**) during the in situ HAADF-STEM observations (5% H_2/Ar , 1 bar, 800 °C). **i**, The particle number as a function of observed time for Ru/SiO_2 and $\text{Ru/LaO}_x\text{-SiO}_2$.

(Fig. 1a). Following high-temperature H_2 treatment, the anions of the second metal are reduced, forming NPs near nano-islands. As shown in Supplementary Fig. 1, inverse pH regulation should be applied for fabricating nano-islands whose metal-containing anions exist steadily in the aqueous solution. Metal cations of supported NPs are suitable for further adsorption on these oxide nano-islands, characterized by a lower IEP value³².

The criteria for manufacturing different categories of oxide nano-islands on normal supports are established (Fig. 1b and Supplementary Fig. 2). Setting the concentration of metal ions as 1 mM enabled the estimation of the pH value corresponding to the K_{sp} of the metal hydroxide and the pH window³³. This library of oxide nano-islands covers 14 representative metal ions, spanning alkali-earth metals, transition metals, *p*-block metals and rare-earth metals. When constructing nano-islands, SiO_2 , TiO_2 and CeO_2 with lower IEP displayed a wide pH window for adsorbing metal cations, whereas Al_2O_3 with higher IEP was suitable for adsorbing metal anions³². A vast number of combinations of catalyst support and oxide nano-island cover most of the metal oxides commonly used as catalyst supports or promoters.

The charge attraction strategy is further employed to create spatial intimacy between oxide nano-islands and metal NPs. Solution pH should be adjusted to realize electrostatic attraction between the active metal ions and nano-islands, while electrostatic repulsion between metal ions and support is preserved (Fig. 1a). Thus, the pH window illustrated in Fig. 1c corresponds to the difference between the IEPs of the oxide support and the nano-island. Metal anions of supported NPs should be chosen for the combination of nano-islands with higher IEP and support, especially SiO_2 and TiO_2 (blue region in Fig. 1c). In contrast, for the combination of Al_2O_3 and nano-islands with low IEP value, metal cations should be chosen (red region in Fig. 1c). The difference in reduction potential between metal ions and nano-islands is used to represent the structural stability of nano-islands³³. When compared with transition metal oxide, the refractory alkali-earth and rare-earth

metal oxides are more suitable to serve as nano-islands (Supplementary Fig. 3). Therefore, we developed a synthetic library to construct ensemble structures for supported catalysts. Using a pH-controlled strong electrostatic adsorption method, metal NPs can be precisely anchored close to the oxide nano-islands.

Among metal ions, La^{3+} was selected as a typical cation to fabricate nano-islands due to its lowest reduction potential, which confers structural stability (Supplementary Fig. 3). SiO_2 with an IEP of 2.8 was further selected to obtain a large pH window for the adsorption of metal anions (Supplementary Fig. 4). Ru was supported to explore the applicability of LaO_x nano-islands in improving sintering resistance, denoted $\text{Ru/LaO}_x\text{-SiO}_2$. To investigate the universality of our method, Mg^{2+} , Al^{3+} and Mn^{2+} cations were also chosen to fabricate nano-islands and support Ru, denoted $\text{Ru/MgO}_x\text{-SiO}_2$, $\text{Ru/AlO}_x\text{-SiO}_2$ and $\text{Ru/MnO}_x\text{-SiO}_2$, respectively. H_2 treatment at 800 °C was performed to ensure the formation of Ru NPs near nano-islands.

Isolation of Ru NPs on nano-islands

Atomic-resolution high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images showed LaO_x species as amorphous and mutually separated clusters on SiO_2 (Fig. 2a and Supplementary Fig. 5). Ru NPs in $\text{Ru/LaO}_x\text{-SiO}_2$ were surrounded by atoms, identified as the edges of LaO_x nano-islands (Fig. 2b and Supplementary Fig. 6). In comparison, for Ru on SiO_2 , denoted Ru/SiO_2 , only bare Ru NPs were observed (Supplementary Fig. 7). Energy dispersive X-ray elemental mapping images showed that most of the Ru and La signals overlapped in the same region for $\text{Ru/LaO}_x\text{-SiO}_2$ (Fig. 2c, d). These results indicate the spatial intimacy between Ru NPs and LaO_x nano-islands. Energy dispersive X-ray elemental mapping images in Supplementary Figs. 8–10 also demonstrate the neighbouring locations between Ru NPs and MgO_x , AlO_x and MnO_x nano-islands, respectively. Thus, the strong electrostatic adsorption method turns out to be a universal approach to fabricating the ensemble structure, where oxide

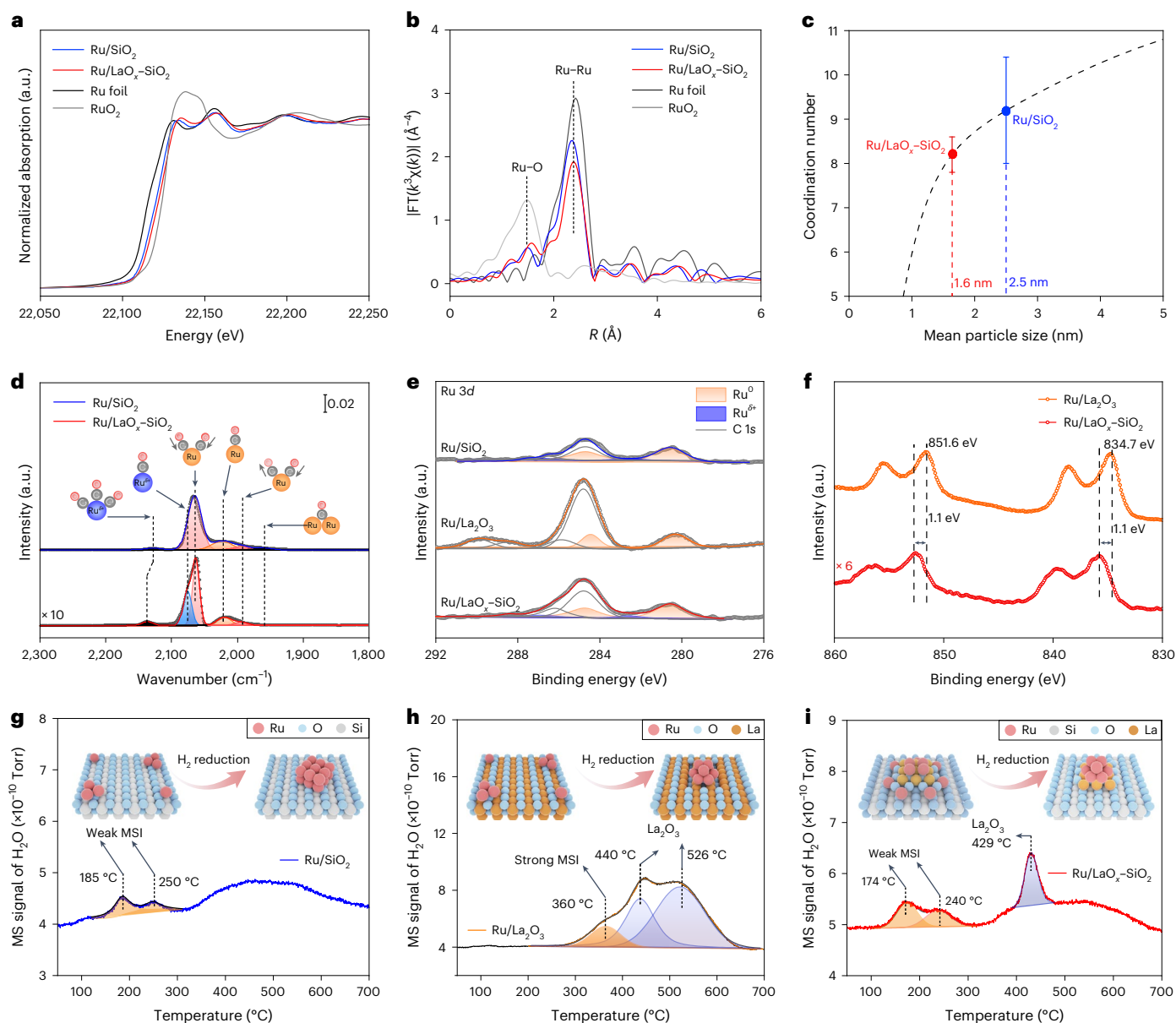


Fig. 3 | Electronic and coordination structure characterizations. a, b, XANES (a) and EXAFS (b) spectra at the Ru K edge of Ru/SiO₂, Ru/LaO_x-SiO₂, Ru foil and RuO₂. a.u., arbitrary units. **c,** Simulated relationship between the coordination number and mean particle size. The data are presented as mean ± s.d. The error bar was introduced during the fitting of EXAFS spectra. **d,** CO-DRIFTS for Ru/SiO₂ and Ru/LaO_x-SiO₂. For clear visibility, the CO-DRIFTS results of

Ru/LaO_x-SiO₂ was enlarged by 10 times. **e,** Ru 3d XPS results for Ru/SiO₂, Ru/La₂O₃ and Ru/LaO_x-SiO₂. **f,** La 3d XPS results for Ru/La₂O₃ and Ru/LaO_x-SiO₂. For clear visibility, the XPS result of Ru/LaO_x-SiO₂ in Fig. 3f was enlarged by 6 times and the spectra in Fig. 3d–f were offset vertically. **g–i,** H₂-TPR profiles of Ru/SiO₂ (g), Ru/La₂O₃ (h) and Ru/LaO_x-SiO₂ (i). The signal of produced water was detected by mass spectrometer (MS).

nano-islands are grafted in between the supported metal NPs and the catalyst support.

For Ru/SiO₂, although a mean size of 2.4 nm was estimated, the presence of NPs around 8 nm suggested the sintering of some Ru NPs (Supplementary Fig. 7). The oxide nano-island helped achieve a more uniform distribution of Ru NPs with smaller sizes (Fig. 2e and Supplementary Fig. 11). For Ru/LaO_x-SiO₂, Ru NPs larger than 2.3 nm were absent, exhibiting a mean size of 1.4 nm, whereas employing bulk oxides rather than nano-islands as support is also futile in preventing Ru sintering. For Ru NPs on MgO, Al₂O₃, MnO₂ and La₂O₃, severe sintering was observed (Supplementary Figs. 12–17). Thus, the above results underscore the necessity of grafting nano-islands in fabricating ultrafine NPs with preserved metal dispersion.

The stabilization effect of different supports on NPs depends on their interfacial adhesion energy (E_{adh})³⁴. This E_{adh} is experimentally

reflected by the contact angle at the interface between metal and oxide, smaller values of which correspond to higher E_{adh} and stronger metal–support interaction (MSI)³⁵. As displayed in high-resolution transmission electron microscopy images, Ru/SiO₂ and Ru/La₂O₃ exhibited mean contact angles of 80° and 40°, respectively (Supplementary Figs. 18–21). Ru/LaO_x-SiO₂ also gave a mean contact angle of 40°, demonstrating that Ru NPs adhered on LaO_x nano-islands (Supplementary Fig. 20). High-resolution transmission electron microscopy images in Supplementary Figs. 22–25 showed that Ru/MgO_x-SiO₂, Ru/AlO_x-SiO₂ and Ru/MnO_x-SiO₂ exhibited mean contact angles of 49°, 72° and 64°, respectively. Ru NPs on LaO_x nano-islands exhibit the smallest contact angle and size, demonstrating that LaO_x exhibited the strongest MSI and stabilization effect on Ru NPs (Supplementary Fig. 26).

The interfacial E_{adh} and chemical potential of Ru were further calculated³⁴. Among different nano-islands, Ru on LaO_x gave the highest

E_{adh} (3.8 J m^{-2} , Supplementary Fig. 27). The chemical potential of a Ru NP on LaO_x was calculated as 99.3 kJ mol^{-1} , much lower than that of Ru on SiO_2 ($160.9 \text{ kJ mol}^{-1}$, Fig. 2f). This huge difference in chemical potential, defined as a potential well, effectively frustrates the sintering of Ru NPs. Under harsh conditions, Ru species migrate on SiO_2 and sinter to minimize chemical potential. For Ru/ $\text{LaO}_x\text{-SiO}_2$, the migrating Ru species are stopped upon contact with LaO_x nano-islands, as the chemical potential is greatly reduced. Moreover, the sintering path of Ru NPs is disconnected by a potential well, as any further extrication would lead to a substantially increased chemical potential. Thus, Ru NPs are isolated and stabilized on LaO_x nano-islands with strong adhesion.

As revealed by in situ HAADF-STEM observations, Ru NPs on SiO_2 migrated vigorously (Fig. 2g and Supplementary Video 1). Initially, five Ru NPs were observed across three distinct regions. During the 200 s observation, the newly emerged Ru NP in region 1 gradually contacted the other Ru NP. Ru NPs in regions 2 and 3 merged to form larger Ru NPs. Conversely, for Ru/ $\text{LaO}_x\text{-SiO}_2$, no aggregation was observed in the whole region (Fig. 2h and Supplementary Video 2). Moreover, for Ru/ SiO_2 , the number of Ru NPs decreased constantly during the 200 s observation, diminishing by one NP per minute (Fig. 2i). In contrast, Ru/ $\text{LaO}_x\text{-SiO}_2$ exhibited solid stability, with the number of Ru NPs remaining constant. Therefore, ultrafine Ru NPs isolated on LaO_x nano-islands exhibit exceptional sintering resistance.

Physicochemical properties of isolated Ru NPs

The quasi-in-situ X-ray absorption near-edge structure (XANES) results in Fig. 3a show that Ru^{4+} precursors in Ru/ SiO_2 and Ru/ $\text{LaO}_x\text{-SiO}_2$ were mostly reduced. Moreover, Ru/ SiO_2 was closer to a metallic state due to its absorption energy being lower than that of Ru/ $\text{LaO}_x\text{-SiO}_2$ ³⁶. According to the extended X-ray absorption fine structure (EXAFS) spectra, Ru–Ru bonds were dominant in both Ru/ SiO_2 and Ru/ $\text{LaO}_x\text{-SiO}_2$, with coordination numbers of 9.2 and 8.2, respectively (Fig. 3b and Supplementary Table 1). The size of the Ru NPs was estimated according to the coordination number³⁷. As shown in Fig. 3c, the sizes of Ru NPs in Ru/ SiO_2 and Ru/ $\text{LaO}_x\text{-SiO}_2$ were deduced to be 2.5 and 1.6 nm, respectively, in close agreement with HAADF-STEM results (2.4 and 1.4 nm).

The CO-probe diffuse reflectance infrared Fourier transform spectroscopy (CO-DRIFTS) measurements in Fig. 3d indicated that characteristic peaks at 2,130, 2,062, 1,990 and 2,020 cm^{-1} were present in both Ru/ SiO_2 and Ru/ $\text{LaO}_x\text{-SiO}_2$. These peaks were assigned to the tricarbonyl species adsorbed on $\text{Ru}^{\delta+}$ ($1 < \delta < 4$), symmetric and asymmetric stretching vibration of dicarbonyl species on Ru^0 and linearly bonded CO on Ru^0 , respectively³⁸. For Ru/ SiO_2 , the bridge-bonded CO at 1,960 cm^{-1} originates from the sintered Ru NPs (Supplementary Fig. 7). For Ru/ $\text{LaO}_x\text{-SiO}_2$, the additional peak at 2,074 cm^{-1} is attributed to the linearly bonded CO on $\text{Ru}^{\delta+}$ at the Ru– LaO_x interface. The great difference in intensity of the above spectra is caused by the rapid desorption of CO in Ru/ $\text{LaO}_x\text{-SiO}_2$ (Supplementary Fig. 28). In short, the CO-DRIFTS results reveal that dominant Ru^0 and minor $\text{Ru}^{\delta+}$ are present in both samples, consistent with the XANES results.

The quasi-in-situ X-ray photoelectron spectroscopy (XPS) results for Ru 3d and 3p showed that Ru^0 dominated in Ru/ SiO_2 and Ru/ $\text{LaO}_x\text{-SiO}_2$, accompanied by minor $\text{Ru}^{\delta+}$ at the Ru–oxide interface (Fig. 3e and Supplementary Fig. 29). Ru/ $\text{LaO}_x\text{-SiO}_2$ exhibited a lower proportion of Ru^0 than did Ru/ SiO_2 due to the abundant Ru– LaO_x interface (Supplementary Fig. 20). For Ru/ La_2O_3 , only Ru^0 existed due to the large size of Ru NPs (Supplementary Fig. 16). The binding energies of La 3d regions for Ru/ $\text{LaO}_x\text{-SiO}_2$ were higher by 1.1 eV than those of Ru/ La_2O_3 , suggesting a lower valence state³⁹ (Fig. 3f). This implies that oxygen vacancies were present in the LaO_x . In contrast to inert SiO_2 , LaO_x nano-islands with oxygen vacancies facilitate electron transfer from Ru to LaO_x clusters, contributing to the higher ratio of $\text{Ru}^{\delta+}$ in Ru/

$\text{LaO}_x\text{-SiO}_2$ ⁴⁰. Moreover, oxygen vacancies in LaO_x may help activate oxygen-containing molecules, such as CO_2 .

H_2 temperature-programmed reduction (H_2 -TPR) experiments were performed to reflect the MSI and in-between E_{adh} (ref. 34). For Ru/ SiO_2 , two peaks at 185 and 250 °C were attributed to the reduction of RuO_x NPs and highly dispersed Ru atoms, respectively (Fig. 3g and Supplementary Fig. 30). For Ru/ La_2O_3 , the elevated reduction temperature of Ru precursors (360 °C) indicates that the E_{adh} of Ru– La_2O_3 is higher than that of Ru– SiO_2 . The pronounced signal of water was ascribed to the reduction of abundant surface oxygen on La_2O_3 near Ru NPs. Furthermore, after Ru was supported, the reduction temperature of La_2O_3 decreased by approximately 100 °C, showing that Ru promoted the reduction of La_2O_3 (Fig. 3h and Supplementary Fig. 31). For Ru/ $\text{LaO}_x\text{-SiO}_2$, the reduction temperatures (174 and 240 °C) were close to those of Ru/ SiO_2 , indicating that most of the Ru precursors previously on SiO_2 were gathered around LaO_x before H_2 reduction (Fig. 3i and Supplementary Fig. 31). Moreover, the reduction peak of LaO_x clusters at 429 °C also implied the presence of oxygen vacancies, in line with XPS results (Fig. 3f). Thus, the above H_2 -TPR results further validate the stabilization of Ru on LaO_x nano-islands, attributed to the higher E_{adh} value of Ru– La_2O_3 when compared with Ru– SiO_2 .

Catalytic performance towards methane dry reforming

The catalytic performance of Ru/ $\text{LaO}_x\text{-SiO}_2$ in the DRM reaction was first evaluated at temperatures between 700 and 800 °C with a gas hourly space velocity (GHSV) of 100 $\text{l g}_{\text{cat}}^{-1} \text{ h}^{-1}$. The conversions of CH_4 and CO_2 increased with temperature and remained steady at each stage (Fig. 4a and Supplementary Fig. 32). The highest conversions and H_2/CO ratio were observed at 800 °C. Furthermore, the GHSV was gradually increased to pursue the highest activity without deactivation. At the GHSV of 250 $\text{l g}_{\text{cat}}^{-1} \text{ h}^{-1}$, Ru/ $\text{LaO}_x\text{-SiO}_2$ exhibited a CH_4 conversion of 83.4%, falling below the equilibrium conversion (91.3%, calculated using HSC 6 software, Supplementary Fig. 33). Further increasing the GHSV to 300 $\text{l g}_{\text{cat}}^{-1} \text{ h}^{-1}$ resulted in slight deactivation (Fig. 4b and Supplementary Fig. 34). Thus, 250 $\text{l g}_{\text{cat}}^{-1} \text{ h}^{-1}$ at 800 °C is chosen for further evaluation.

The time-on-stream stability test is shown in Fig. 4c and Supplementary Fig. 35. Ru/ La_2O_3 experienced pronounced deactivation during the 40 h test, as evidenced by a rapid decrease in CH_4 conversion from 79% to 8% and H_2/CO ratio from 0.85 to 0.21. Ru/ SiO_2 exhibited an initial CH_4 conversion of 89% and H_2/CO ratio of 0.96. For Ru/ $\text{LaO}_x\text{-SiO}_2$, the slightly lower initial conversion, 83%, was attributed to the higher valence state (Fig. 3e). During the 150 h test, the CH_4 conversion of Ru/ SiO_2 dropped to 38%, accompanied by a decrease in H_2/CO ratio to 0.76. Conversely, Ru/ $\text{LaO}_x\text{-SiO}_2$ exhibited a steady CH_4 conversion of around 83% and H_2/CO of 0.94. Negligible deactivation was observed during the 400 h test, attributed to the stabilization effect of LaO_x nano-islands (Fig. 2f). In the DRM reaction, the lack of catalytic stability is usually compensated by sacrificing active sites, known as the activity–stability trade-off. This is particularly normal in the metal encapsulation strategy, as the immobilized NPs were physically covered. For Ru/ $\text{LaO}_x\text{-SiO}_2$, this improvement achieved through regulating electronic properties preserves the exposure of active sites to the utmost, reaching solid stability at slight cost to the activity.

To quantitatively assess the catalytic stability, an indicator of activity maintenance was defined:

$$\text{activity maintenance} = (X_{\text{CH}_4, \text{final}}/X_{\text{CH}_4, \text{initial}}) \times 100\% \quad (1)$$

where $X_{\text{CH}_4, \text{final}}$ and $X_{\text{CH}_4, \text{initial}}$ represent the final and initial CH_4 conversions, respectively. As a result, among normal rare-earth oxides, the Ru on LaO_x nano-islands exhibited the highest activity maintenance

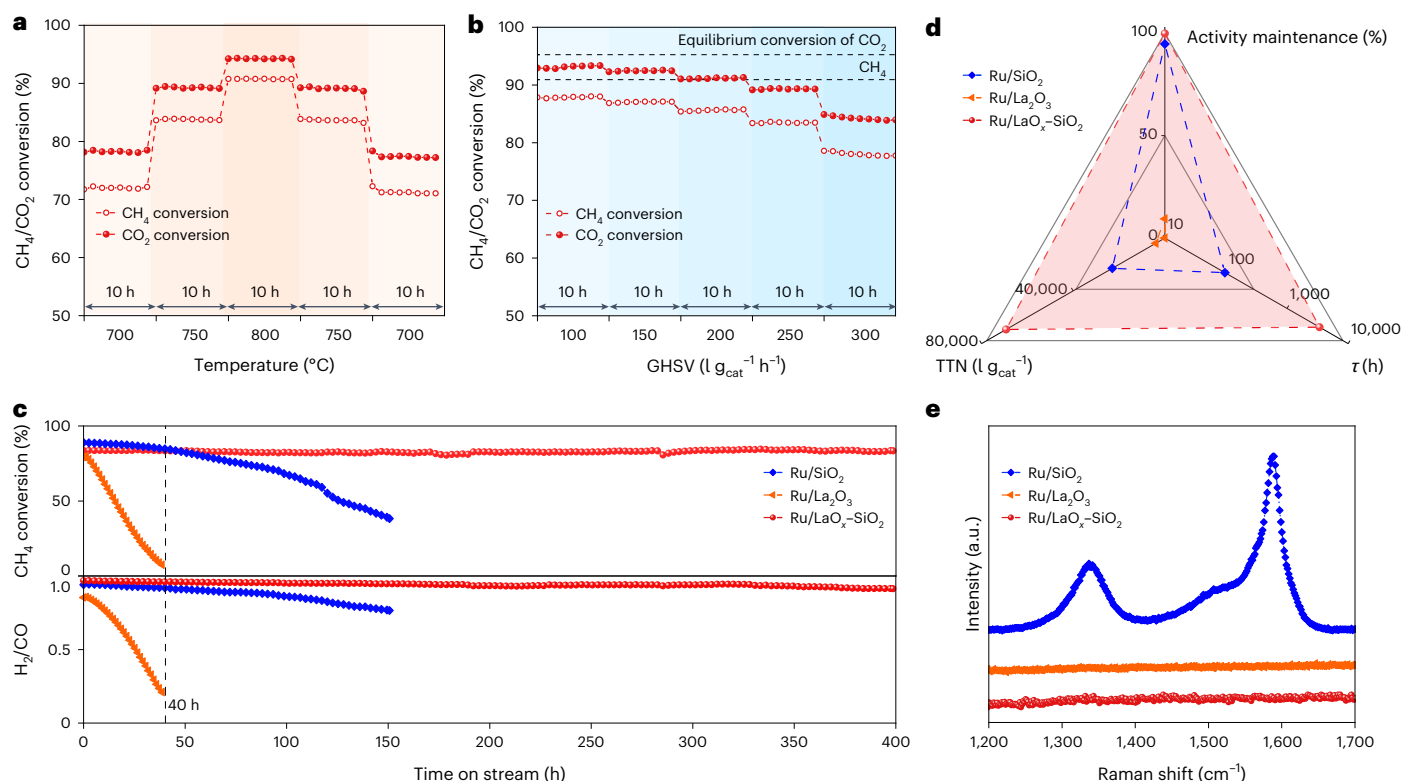


Fig. 4 | Catalytic performance for DRM reaction. **a**, DRM activity over Ru/LaO_x-SiO₂ at different temperatures with the GHSV of 100 l g_{cat}⁻¹ h⁻¹. **b**, DRM activity over Ru/LaO_x-SiO₂ at 800 °C with different GHSVs. **c**, The DRM stability test (800 °C, 1 bar, 250 l g_{cat}⁻¹ h⁻¹, CH₄:CO₂:N₂ = 2:2:1). **d**, Catalytic performance

comparison of the Ru/SiO₂, Ru/La₂O₃ and Ru/LaO_x-SiO₂. TTN, total turnover number. **e**, Raman spectra of the Ru/SiO₂, Ru/La₂O₃ and Ru/LaO_x-SiO₂ after the DRM stability test.

(Supplementary Fig. 36). The mean lifetime (τ) of catalysts was also calculated according to

$$\tau = t / \{ \ln[(1 - X_{\text{CH}_4, \text{final}}) / X_{\text{CH}_4, \text{final}}] - \ln[(1 - X_{\text{CH}_4, \text{initial}}) / X_{\text{CH}_4, \text{initial}}] \} \quad (2)$$

where t represents the reaction time. The τ and activity maintenance on the basis of the 40 h operation, as well as the total turnover number during the stability test, were calculated and compared (Fig. 4d and Supplementary Table 2). Ru/LaO_x-SiO₂ exhibited a τ of 3,403 h, activity maintenance of 99.8% and total turnover number of 69,250 l_{reactant} g_{cat}⁻¹, substantially outperforming Ru/SiO₂ and Ru/La₂O₃⁴¹. LaO_x-SiO₂ and La₂O₃ were almost inert toward the DRM reaction (Supplementary Figs. 37 and 38). As revealed by Raman spectra, coke formation was observed on Ru/SiO₂, while carbon signal was absent for Ru/La₂O₃ and Ru/LaO_x-SiO₂ (Fig. 4e).

Catalyst durability benefiting from sintering resistance

For the spent Ru/La₂O₃ (after the DRM stability test), HAADF-STEM images indicated that the mean size of Ru NPs increased from 12.4 to 16.3 nm (Supplementary Figs. 16 and 39). The severe sintering was considered the cause of its rapid deactivation. Moreover, the phase change from La₂O₃ to La₂O₂CO₃ also exacerbated the catalyst deactivation (Supplementary Figs. 40–42). For spent Ru/SiO₂, the mean size increased from 2.4 to 10.9 nm, indicating severe sintering (Supplementary Fig. 7 and Fig. 5a). Furthermore, instead of coke formation, sintering was demonstrated to be responsible for the deactivation of Ru/SiO₂ (Supplementary Figs. 43–47). Conversely, for spent Ru/LaO_x-SiO₂, the mean size of Ru NPs was still 1.4 nm with an absence of Ru NPs larger than 2.5 nm (Fig. 5b). The preservation of Ru sites on LaO_x was further

demonstrated through CO pulse experiments (Supplementary Fig. 48). Thus, compared with the severely sintered Ru/SiO₂ and Ru/La₂O₃, the sintering propensity on Ru/LaO_x-SiO₂ was inhibited.

The sintering process of Ru/SiO₂ and Ru/LaO_x-SiO₂ was monitored by reducing samples at 800 °C for different periods. X-ray diffraction (XRD) patterns were recorded for these reduced samples (Fig. 5c). For Ru/SiO₂, characteristic peaks of Ru facets gradually emerged and became prominent, suggesting the ongoing sintering. In contrast, for Ru/LaO_x-SiO₂, the hump around 44°, pointing to (101) and (002) facets, remained unchanged throughout the reduction process. After DRM reaction for different times, a trend in XRD patterns similar to those after H₂ reduction was observed (Supplementary Fig. 49). For Ru on MgO_x, AlO_x and MnO_x nano-islands, XRD patterns indicated that Ru NPs maintained their size during H₂ reduction (Supplementary Fig. 50). The above results indicate that the ultrafine Ru NPs isolated on nano-islands exhibit exceptional sintering resistance.

Coke formation, arising from mismatched CH₄ and CO₂ activation, is another notorious factor causing the low durability of DRM catalysts⁴². As revealed by the CH₄ temperature-programmed surface reaction experiments, for Ru/LaO_x-SiO₂ the initial temperature of CH₄ dissociation was 249 °C, slightly higher than that of Ru/SiO₂ (236 °C, Fig. 5d). CH₄-D₂ exchange experiments indicated a more intense HD and CH₃D signal on Ru/SiO₂ (Supplementary Fig. 51). The above results suggest that Ru/LaO_x-SiO₂ exhibited weaker CH₄ activation, attributed to the higher valence state (Fig. 3a). This observation agrees with the XPS results and catalytic activities (Figs. 3e and 4d).

Temperature-dependent CO₂-DRIFTS results suggested that, for Ru/SiO₂, the adsorption peak arising from CO₂ dissociation was absent at 250 °C (Fig. 5e). At 300 °C, a weak adsorption peak of linearly bonded and bridge-bonded CO was observed. In contrast, Ru/LaO_x-SiO₂ exhibited a prominent peak of linearly adsorbed CO at 250 °C. The facilitation

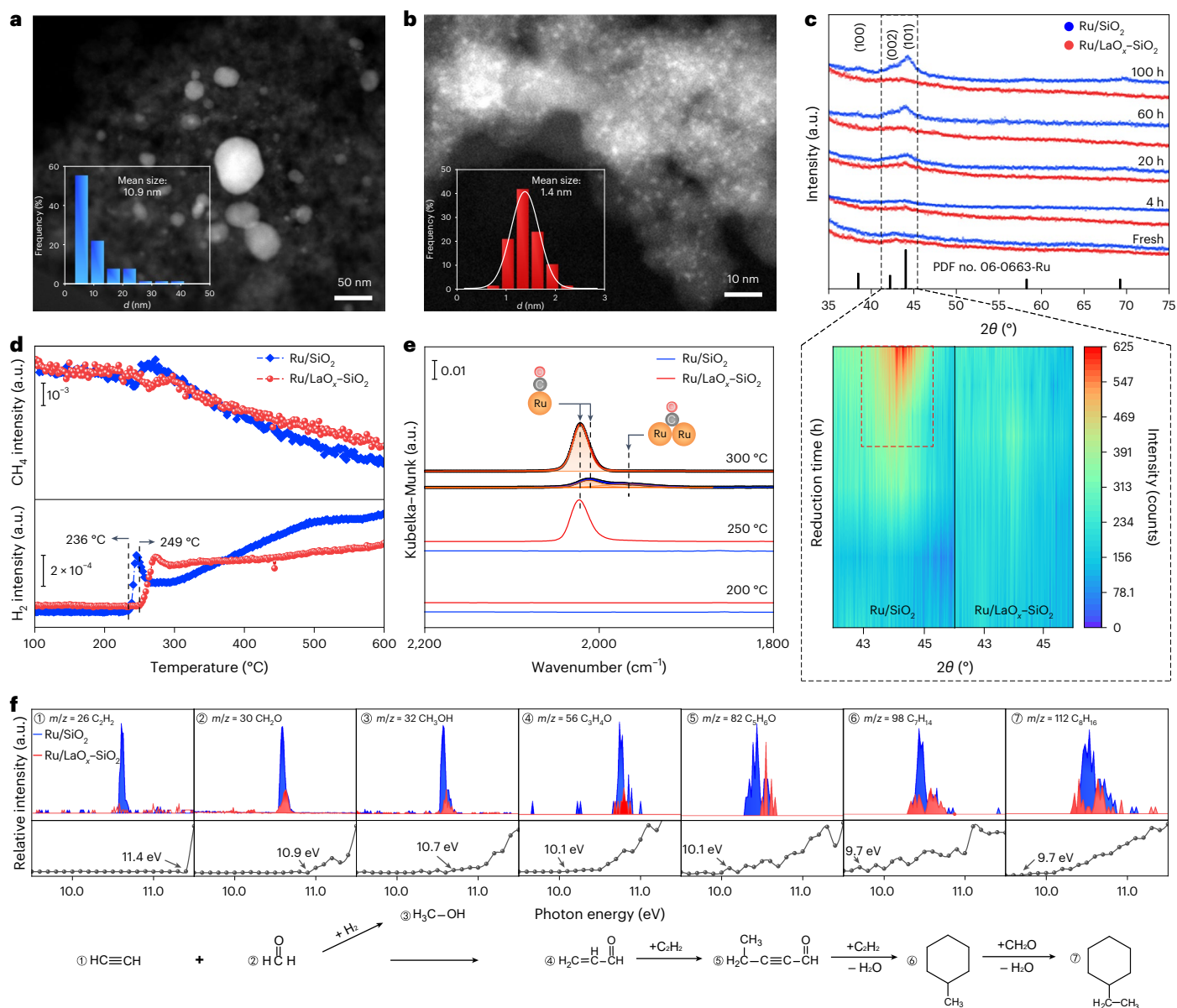


Fig. 5 | Investigation of catalyst durability. **a, b**, HAADF-STEM images of spent Ru/SiO₂ (**a**) and Ru/LaO_x-SiO₂ (**b**). **c**, XRD patterns and the corresponding contour plots of Ru/SiO₂ and Ru/LaO_x-SiO₂ after H₂ reduction at 800 °C for different times. PDF, Powder Diffraction File. **d**, CH₄ temperature-programmed surface

reaction spectra of Ru/SiO₂ and Ru/LaO_x-SiO₂. **e**, CO₂-DRIFTS spectra of Ru/SiO₂ and Ru/LaO_x-SiO₂ at 200 °C, 250 °C and 300 °C. **f**, SVUV-PIMS spectra of reaction intermediates and the proposed mechanism of coke formation during the DRM reaction. Reaction conditions: 2 Torr, 800 °C, 14.5 eV, CH₄:CO₂:N₂ = 2:2:1.

of CO₂ activation by LaO_x was further proved by density functional theory calculations (Supplementary Figs. 52 and 53). To directly access reactant activation at 800 °C, CH₄ and CO₂ were introduced stepwise into Ru/SiO₂ and Ru/LaO_x-SiO₂. As a result, for Ru/LaO_x-SiO₂, higher intensity of unconverted CH₄ and lower intensity of produced CO pointed to weaker CH₄ activation but facilitated CO₂ activation (Supplementary Fig. 54). Thus, LaO_x nano-islands balanced the mismatched CH₄ and CO₂ activation, benefiting the inhibition of coke formation.

The mechanism of coke formation was unveiled by using synchrotron-based vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). As shown in Fig. 5f and Supplementary Fig. 55, a series of reaction intermediates, including acetylene, formaldehyde, methanol, 2-propenal, 2-pentynal, methylcyclohexane and ethylcyclohexane, was observed during the DRM reaction. A mechanism of coke formation was thus proposed. As initiators of coke formation, formaldehyde reacts with acetylene to form α, β-unsaturated species, such as 2-propenal and 2-pentynal^{43,44}. These unsaturated species

continue to react with acetylene and formaldehyde, undergoing dehydration and cyclization to form cyclic hydrocarbons. Further dehydrogenation and aromatization generate polycyclic aromatic species, eventually evolving into coke⁴⁵. Noteworthy, the intensity of reaction intermediates for Ru/LaO_x-SiO₂ was much lower than that for Ru/SiO₂. Thus, LaO_x nano-islands enhance the coke resistance of Ru NPs by inhibiting the formation of reactant intermediates.

The broad applicability of our isolation strategy was further investigated. As revealed by Supplementary Figs. 56–60, the sintering resistance and catalytic performance in DRM of Ir and Pt NPs were greatly improved through isolating NPs on LaO_x. Moreover, the strategy has proved effective using a carbon support (Supplementary Figs. 61–63). Thus, our isolation strategy is demonstrated as a universal method for improving sintering resistance, applicable to many noble metals on various supports. More broadly, through customizing the active metals and nano-islands, we anticipate that this structure will prove useful for further catalytic transformations.

Under harsh reaction conditions, the sintering of NPs is driven by minimizing surface energy. An isolation strategy is developed to eliminate the sintering propensity of ultrafine NPs by reducing surface energy via strong adhesion of nano-islands. A synthetic library has been established, among which the LaO_x nano-island is the most effective stabilizer, facilitating the localization of Ru NPs. Afterwards, the thermodynamic negativity to liberate NPs from strong adhesion disconnects the path of further metal migration. Following this isolation strategy, we fabricated a $\text{Ru/LaO}_x\text{-SiO}_2$ catalyst, which exhibited a steady CH_4 conversion rate of around $1,040 \mu\text{mol}_{\text{CH}_4} \text{g}_{\text{cat}}^{-1} \text{s}^{-1}$ during the 400 h DRM test. Noteworthily, ultrafine Ru NPs in $\text{Ru/LaO}_x\text{-SiO}_2$ maintain a mean size of 1.4 nm after the long-term DRM test. When compared with common strategies such as alloying or encapsulation^{31,46,47}, the isolation strategy achieves exceptional sintering resistance under harsh reaction conditions without the involvement of high-cost or scarce reagents, delicate instruments or sophisticated synthetic processes. The easily scalable fabrication of nano-islands shows potential for industrial applications.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41563-025-02134-9>.

References

- Mitchell, M. J. et al. Engineering precision nanoparticles for drug delivery. *Nat. Rev. Drug Discov.* **20**, 101–124 (2020).
- Li, J.-F., Zhang, Y.-J., Ding, S.-Y., Panneerselvam, R. & Tian, Z.-Q. Core-shell nanoparticle-enhanced Raman spectroscopy. *Chem. Rev.* **117**, 5002–5069 (2017).
- Lu, Z. & Yin, Y. Colloidal nanoparticle clusters: functional materials by design. *Chem. Soc. Rev.* **41**, 6874–6887 (2012).
- Liu, Y., Zhao, G., Wang, D. & Li, Y. Heterogeneous catalysis for green chemistry based on nanocrystals. *Natl Sci. Rev.* **2**, 150–166 (2015).
- Xie, C., Niu, Z., Kim, D., Li, M. & Yang, P. Surface and interface control in nanoparticle catalysis. *Chem. Rev.* **120**, 1184–1249 (2020).
- Ding, K. et al. A general synthesis approach for supported bimetallic nanoparticles via surface inorganometallic chemistry. *Science* **362**, 560–564 (2018).
- Ji, S., Jun, C., Chen, Y. & Wang, D. Precise synthesis at the atomic scale. *Precis. Chem.* **1**, 199–225 (2023).
- Dai, Y., Lu, P., Cao, S., Campbell, C. T. & Xia, Y. The physical chemistry and materials science behind sinter-resistant catalysts. *Chem. Soc. Rev.* **47**, 4314–4331 (2018).
- Wu, S., Li, M. & Sun, Y. *In situ* synchrotron X-ray characterization shining light on the nucleation and growth kinetics of colloidal nanoparticles. *Angew. Chem. Int. Ed.* **58**, 8987–8995 (2019).
- Wiebenga, M. H. et al. Deactivation mechanisms of Pt/Pd-based diesel oxidation catalysts. *Catal. Today* **184**, 197–204 (2012).
- Baudouin, D. et al. Particle size effect in the low temperature reforming of methane by carbon dioxide on silica-supported Ni nanoparticles. *J. Catal.* **297**, 27–34 (2013).
- Malte, H. The DLVO theory in microbial adhesion. *Colloids Surf. B* **14**, 105–119 (1999).
- Zhang, W. Nanoparticle aggregation: principles and modeling. *Nanomaterials* **811**, 19–43 (2014).
- Parkinson, G. S. et al. Carbon monoxide-induced adatom sintering in a $\text{Pd-Fe}_3\text{O}_4$ model catalyst. *Nat. Mater.* **12**, 724–728 (2013).
- Martín, A. J., Mitchell, S., Mondelli, C., Jaydev, S. & Pérez-Ramírez, J. Unifying views on catalyst deactivation. *Nat. Catal.* **5**, 854–866 (2022).
- Hansen, T. W., DeLaRiva, A. T., Challa, S. R. & Datye, A. K. Sintering of catalytic nanoparticles: particle migration or Ostwald ripening? *Acc. Chem. Res.* **46**, 1720–1730 (2013).
- Goodman, E. D., Schwalbe, J. A. & Cargnello, M. Mechanistic understanding and the rational design of sinter-resistant heterogeneous catalysts. *ACS Catal.* **7**, 7156–7173 (2017).
- Challa, S. R. et al. Relating rates of catalyst sintering to the disappearance of individual nanoparticles during Ostwald ripening. *J. Am. Chem. Soc.* **133**, 20672–20675 (2011).
- Yin, P. et al. Quantification of critical particle distance for mitigating catalyst sintering. *Nat. Commun.* **12**, 4865 (2021).
- Farmer, J. A. & Campbell, C. T. Ceria maintains smaller metal catalyst particles by strong metal-support bonding. *Science* **329**, 933–936 (2010).
- Joo, S. H. et al. Thermally stable Pt/mesoporous silica core-shell nanocatalysts for high-temperature reactions. *Nat. Mater.* **8**, 126–131 (2009).
- Liu, Q. et al. Sintering behavior of carbon-supported Pt nanoparticles and the effect of surface overcoating. *Mater. Today Nano* **20**, 100273 (2022).
- Wang, L., Wang, L., Meng, X. & Xiao, F.-S. New strategies for the preparation of sinter-resistant metal-nanoparticle-based catalysts. *Adv. Mater.* **31**, 1901905 (2019).
- Lu, J. et al. Coking- and sintering-resistant palladium catalysts achieved through atomic layer deposition. *Science* **335**, 1205–1208 (2012).
- Peng, H. et al. Catalysts in coronas: a surface spatial confinement strategy for high-performance catalysts in methane dry reforming. *ACS Catal.* **9**, 9072–9080 (2019).
- Aitbekova, A. et al. Templated encapsulation of platinum-based catalysts promotes high-temperature stability to 1,100 °C. *Nat. Mater.* **21**, 1290–1297 (2022).
- Baker, R. T. K. The relationship between particle motion on a graphite surface and Tammann temperature. *J. Catal.* **78**, 473–476 (1982).
- Pakhare, D. & Spivey, J. A review of dry (CO_2) reforming of methane over noble metal catalysts. *Chem. Soc. Rev.* **43**, 7813–7837 (2014).
- Kim, S. M. et al. Cooperativity and dynamics increase the performance of NiFe dry reforming catalysts. *J. Am. Chem. Soc.* **139**, 1937–1949 (2017).
- Li, X. et al. Functional CeO_x nanoglues for robust atomically dispersed catalysts. *Nature* **611**, 284–288 (2022).
- Wong, A., Liu, Q., Griffin, S., Nicholls, A. & Regalbuto, J. R. Synthesis of ultrasmall, homogeneously alloyed, bimetallic nanoparticles on silica supports. *Science* **358**, 1427–1430 (2017).
- Kosmulski, M. Isoelectric points and points of zero charge of metal (hydr)oxides: 50 years after Parks' review. *Adv. Colloid Interface Sci.* **238**, 1–61 (2016).
- Rumble, J. R. *CRC Handbook of Chemistry and Physics* (CRC Press/Taylor & Francis, 2021).
- Hu, S. & Li, W.-X. Sabatier principle of metal-support interaction for design of ultrastable metal nanocatalysts. *Science* **374**, 1360–1365 (2021).
- Liu, J.-C. et al. Metal affinity of support dictates sintering of gold catalysts. *J. Am. Chem. Soc.* **144**, 20601–20609 (2022).
- Peng, X. et al. Size-dependent activity of supported Ru catalysts for ammonia synthesis at mild conditions. *J. Catal.* **408**, 98–108 (2022).
- Jentys, A. Estimation of mean size and shape of small metal particles by EXAFS. *Phys. Chem. Chem. Phys.* **1**, 4059–4063 (1999).

38. Chin, S. Y., Williams, C. T. & Amiridis, M. D. FTIR studies of CO adsorption on Al_2O_3 - and SiO_2 -supported Ru catalysts. *J. Phys. Chem. B* **110**, 871–882 (2006).
39. John, F. M. et al. *Handbook of X-ray Photoelectron Spectroscopy* (Perkin-Elmer, 1992).
40. Pacchioni, G. Electronic interactions and charge transfers of metal atoms and clusters on oxide surfaces. *Phys. Chem. Chem. Phys.* **15**, 1737–1757 (2013).
41. Song, Y. et al. Response to comment on “Dry reforming of methane by stable Ni–Mo nanocatalysts on single-crystalline MgO ”. *Science* **368**, eabb5680 (2020).
42. Li, H., Hao, C., Tian, J., Wang, S. & Zhao, C. Ultra-durable Ni–Ir/ MgAl_2O_4 catalysts for dry reforming of methane enabled by dynamic balance between carbon deposition and elimination. *Chem Catal.* **2**, 1748–1763 (2022).
43. Luo, J., Xiao, T., Liu, C. & Pan, Y. Recent progress on the involvement of formaldehyde in the methanol-to-hydrocarbons reaction. *ChemSusChem* **18**, e202400884 (2024).
44. Çağlayan, M. et al. Initial carbon–carbon bond formation during the early stages of methane dehydroaromatization. *Angew. Chem. Int. Ed.* **59**, 16741–16746 (2020).
45. Zhang, Q. et al. Deciphering transient carbon footprint for methane dry reforming over Ni@S-1 nanosheet zeolite catalyst. *Appl. Catal. B* **362**, 124722 (2025).
46. Yan, H. et al. Tandem In_2O_3 –Pt/ Al_2O_3 catalyst for coupling of propane dehydrogenation to selective H_2 combustion. *Science* **371**, 1257–1260 (2021).
47. Zhang, J. et al. Sinter-resistant metal nanoparticle catalysts achieved by immobilization within zeolite crystals via seed-directed growth. *Nat. Catal.* **1**, 540–546 (2018).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2025

Methods

Preparation of $\text{LaO}_x\text{-SiO}_2$

In a typical synthesis of $\text{LaO}_x\text{-SiO}_2$, 360 mg of SiO_2 was dispersed in 100 ml of deionized water at room temperature. The pH of the solution was adjusted to 8–9 by adding 1 ml of 2 M ammonia solution. The solution was stirred for 10 min to ensure a negatively charged SiO_2 surface. Subsequently, 45.0 mg of $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ was poured into the solution. The solution was stirred for an additional 10 min. Vacuum filtration was applied to separate the sample, which was dried at room temperature for 3 d. The loading of La on SiO_2 was determined as 3.3 wt% by inductively coupled plasma–atomic emission spectroscopy measurements.

Preparation of $\text{Ru/LaO}_x\text{-SiO}_2$

200 mg of $\text{LaO}_x\text{-SiO}_2$ was dispersed in 50 ml of deionized water at room temperature. The pH of the solution was controlled around 3 by adding 20 μl of HCl aqueous solution. In a typical synthesis, 17.3 mg of $(\text{NH}_4)_2\text{RuCl}_6$ was dissolved in 25 ml of deionized water to obtain the precursor solution (pH 3 with the addition of HCl aqueous solution). With vigorous stirring, 20 ml of $(\text{NH}_4)_2\text{RuCl}_6$ solution was injected into $\text{LaO}_x\text{-SiO}_2$ solution in 30 min. Subsequently, rotary evaporation at 70 °C under 30 mbar was utilized to recover the $\text{Ru/LaO}_x\text{-SiO}_2$ catalyst, which was further dried at room temperature for 3 d. The sample was reduced under 10% H_2/Ar at 800 °C for 1 h before the DRM test.

Preparation of Ru/SiO_2

The synthetic procedure of Ru/SiO_2 , with loading of 2 wt%, was the same as that of $\text{Ru/LaO}_x\text{-SiO}_2$ except for changing the support from $\text{LaO}_x\text{-SiO}_2$ to SiO_2 .

Catalytic reaction

The DRM test was carried out in a fixed-bed reactor under atmospheric pressure. 15 mg of catalyst diluted with 985 mg of quartz sand was packed inside the quartz tube with an internal diameter of 9 mm. The catalyst was reduced in 10% H_2/Ar at 800 °C for 1 h. Subsequently, the temperature was kept at 800 °C or decreased to 750 °C or 700 °C. The reaction was operated in mixed gas ($\text{CH}_4:\text{CO}_2:\text{N}_2 = 2:2:1$) at a pressure of 1 bar. The GHSV was controlled within 100–300 $\text{l g}_{\text{cat}}^{-1} \text{h}^{-1}$. Effluent gas was analysed by a gas chromatograph (Shimadzu) equipped with a TDX-01 column and a thermal conductivity detector, using Ar as carrier gas. Conversions of CH_4 and CO_2 were calculated according to equations (3) and (4). The carbon balance in this work was between 95% and 105%.

$$X_{\text{CH}_4}(\%) = 100 \times \left\{ \frac{[\text{CH}_4]_{\text{inlet}}/[\text{N}_2]_{\text{inlet}} - [\text{CH}_4]_{\text{outlet}}/[\text{N}_2]_{\text{outlet}}}{[\text{CH}_4]_{\text{inlet}}/[\text{N}_2]_{\text{inlet}}} \right\} \quad (3)$$

$$X_{\text{CO}_2}(\%) = 100 \times \left\{ \frac{[\text{CO}_2]_{\text{inlet}}/[\text{N}_2]_{\text{inlet}} - [\text{CO}_2]_{\text{outlet}}/[\text{N}_2]_{\text{outlet}}}{[\text{CO}_2]_{\text{inlet}}/[\text{N}_2]_{\text{inlet}}} \right\} \quad (4)$$

where $[\text{CH}_4]_{\text{inlet}}$, $[\text{CO}_2]_{\text{inlet}}$ and $[\text{N}_2]_{\text{inlet}}$ are the integral areas in the chromatography of CH_4 , CO_2 and N_2 in the feed stream of the inlet, respectively. $[\text{CH}_4]_{\text{outlet}}$, $[\text{CO}_2]_{\text{outlet}}$ and $[\text{N}_2]_{\text{outlet}}$ are the integral areas in the chromatography of CH_4 , CO_2 and N_2 in the feed stream of the outlet, respectively. The concentrations of H_2 and CO are determined by an external standard method.

In situ HAADF-STEM observations

In situ HAADF-STEM observations were conducted on a JEOL JEM-F200 microscope with a beam current of less than 30 pA. The sample was dispersed in ethanol by sonication. The suspension was deposited directly onto a thermal E-chip. The equipped thin heating membrane was controlled by a Protochips atmosphere system. Another E-chip window was placed on top of the thermal chip in the sample holder to create a gas cavity sealed from the high vacuum of the TEM column. Samples were situated between two SiN membranes, each 30–50 nm

in thickness with a 5 μm gap. The real-time observations were conducted during both the heating and constant-temperature processes. The sample was heated to 500 °C with a ramping rate of 180 °C min^{-1} . Subsequently, the sample was further heated to 800 °C with a ramping rate of 30 °C min^{-1} . The sample was always in 5% H_2/Ar with a flow rate of 0.1 ml min^{-1} . The pressure was maintained at 1 bar to closely simulate the real reduction process. The in situ HAADF-STEM data were processed using Protochips AXON studio.

SVUV-PIMS measurements

SVUV-PIMS measurements were carried out at the combustion beamline (BL03U) of the National Synchrotron Radiation Laboratory at Hefei, China. Each catalyst was packed in a quartz reactor with a nozzle size of approximately 0.1 mm. The reactor was connected to an online SVUV-PIMS spectrometer. Catalysts were first reduced in H_2 flow ($\text{H}_2:\text{Ar} = 1:9$) with a total flow rate of 40 ml min^{-1} at 800 °C for 1 h. Subsequently, the gas was switched to reactive gas ($\text{CH}_4:\text{CO}_2:\text{Ar} = 2:2:1$) and data were acquired simultaneously. The photon energy was controlled at 14.5 eV and pressure was controlled at 2 Torr. To identify specific species, we changed the ionization energy from 9.5 eV to 11.5 eV at intervals of 0.1 eV during the DRM reaction over Ru/SiO_2 .

Data availability

All data supporting the findings of this study are available within the main text, figures and Supplementary Information, or from the corresponding authors upon request. Source data are provided with this paper.

Acknowledgements

J. Zeng acknowledges the National Key Research and Development Program of China (2021YFA1500500), CAS Project for Young Scientists in Basic Research (YSBR-051), National Science Fund for Distinguished Young Scholars (21925204), NSFC (2221003, 22250007, 22361162655), Fundamental Research Funds for the Central Universities, Joint Fund of Yulin University and the Dalian National Laboratory for Clean Energy (YLU-DNL Fund 2022012), State Key Laboratory of Catalysis (2024SKL-A-01), International Partnership Program of Chinese Academy of Sciences (123GJHZ2022101GC) and the New Cornerstone Science Foundation through the XPLOER PRIZE. H.Y. acknowledges the NSFC (22302185), the Fundamental Research Funds for the Central Universities (WK9990000167), Young Elite Scientists Sponsorship Program by CAST (2022QNR0001) and Anhui Natural Science Foundation for Young Scholars (2308085QB53). X.L. acknowledges the NSFC (22201272) and Central Government Funds of Guiding Local Scientific and Technological Development for Sichuan Province (2024ZYD0152). H.L. acknowledges the NSFC (22308346) and the Science Fund for Distinguished Young Scholars of Anhui Province (2408085J008). Y.W. acknowledges the NSFC (22405257). L.L. acknowledges the Natural Science Foundation of Anhui Province (2208085QB48). This work was partially carried out at the Instruments Center for Physical Science, University of Science and Technology of China. We thank the BL14W1 beamline at Shanghai Synchrotron Radiation Facility for the XAFS measurements. W.-X. Li (USTC) and H. Zhang (NKU) are acknowledged for their valuable discussions.

Author contributions

T.Z., H.Y. and J. Zeng came up with the idea and designed the study. T.Z., X.L. and L.L. synthesized catalysts and conducted structural characterizations. T.Z. and Z. Zhao conducted the catalytic performance evaluation. Y.W. and T.Z. conducted the in situ HAADF-STEM experiments. J. Zhao conducted DFT calculations. S.H. conducted the high-resolution transmission electron microscopy characterization. R.W. processed the XAFS results. T.Z., Z.X., C.L. and L.Z. conducted the SVUV-PIMS experiment. W.W., H.L. and Z. Zhang provided valuable suggestions on the research project. T.Z. and H.Y.

analysed the data and co-wrote the paper. J. Zeng supervised the work. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41563-025-02134-9>.

Correspondence and requests for materials should be addressed to Han Yan or Jie Zeng.

Peer review information *Nature Materials* thanks the anonymous reviewers for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.